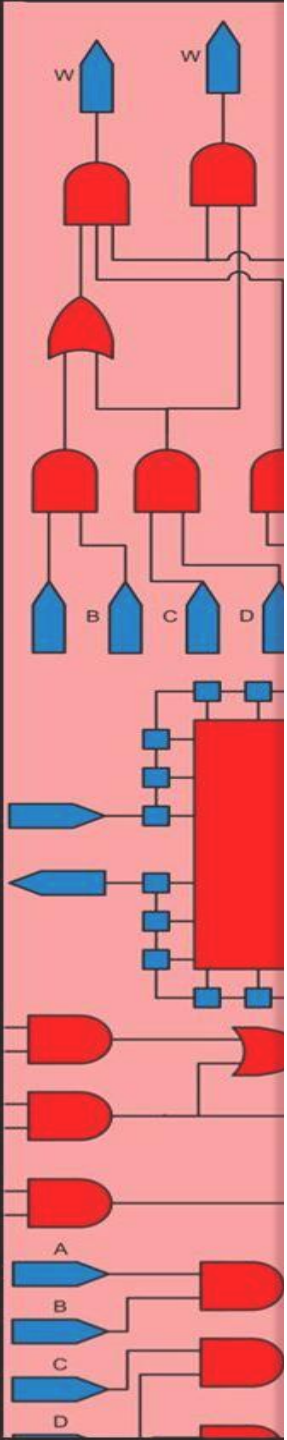
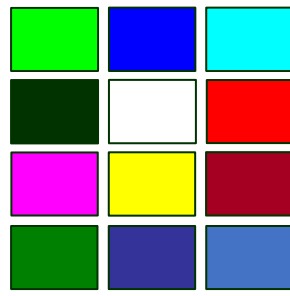
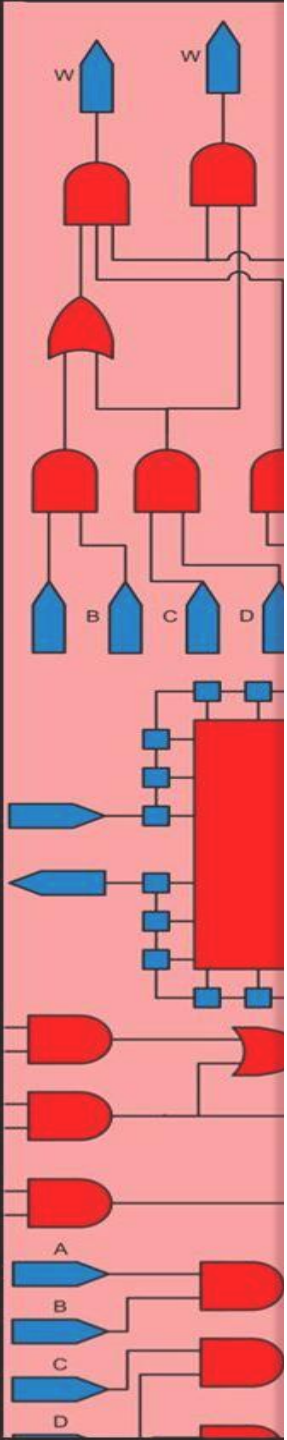


Topic 6

RTL Communication





Outline

Intra – RTL Communication

Multiplexed Bus

Three State Bus

Bidirectional Bus

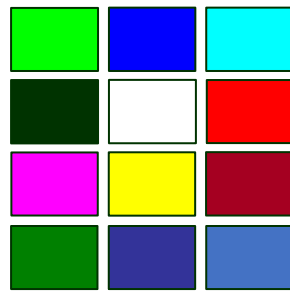
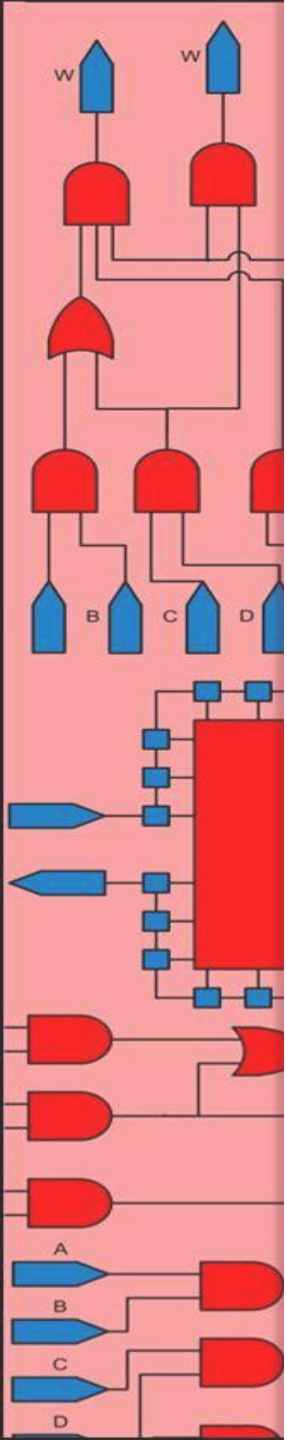
Inter – RTL Communication

What is handshaking?

Device-to-device handshaking

Memory handshaking

IO wrappers



Outline

Intra – RTL Communication

Multiplexed Bus

Three State Bus

Bidirectional Bus

Inter – RTL Communication

What is handshaking?

Device-to-device handshaking

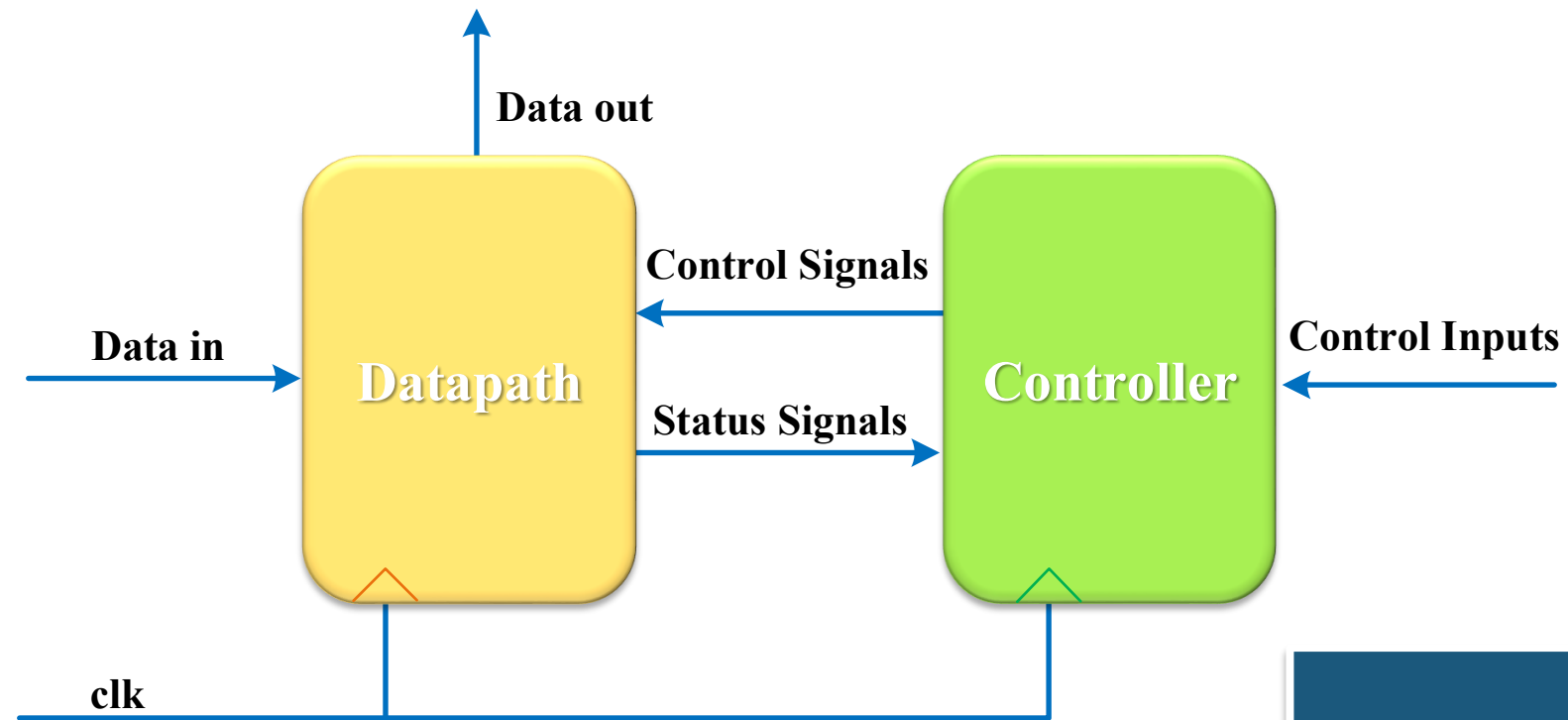
Memory handshaking

Bus Arbitration

IO wrappers

Intra – RTL Communication

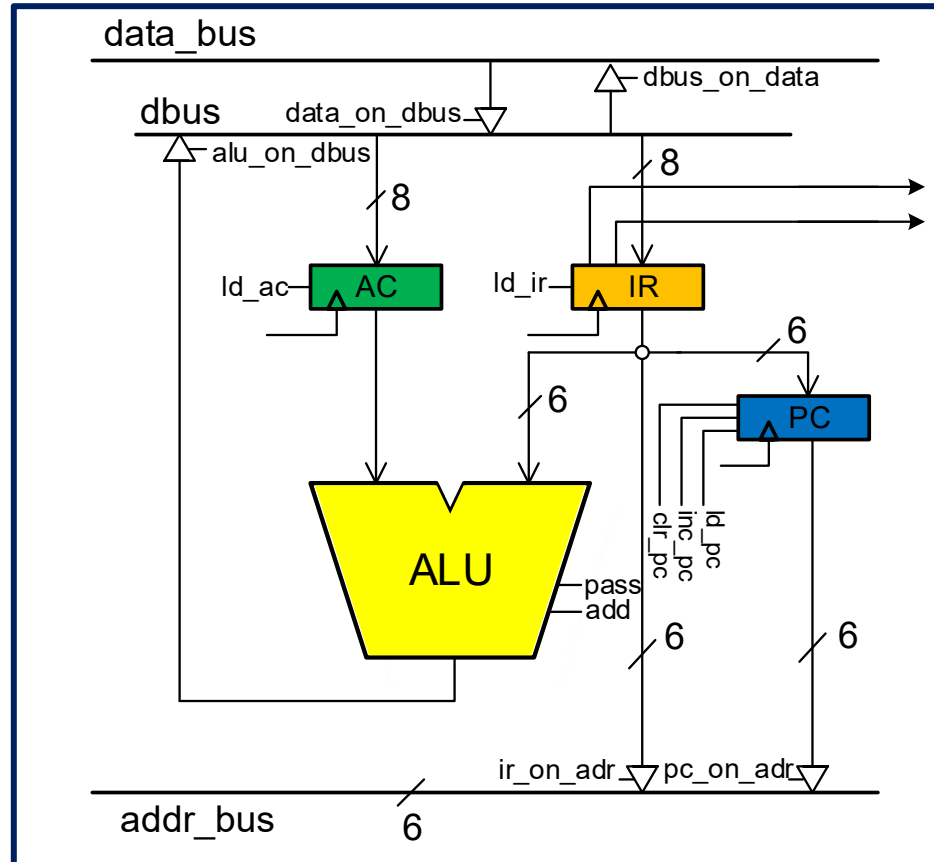
- We have learned intra-RTL communication so far.



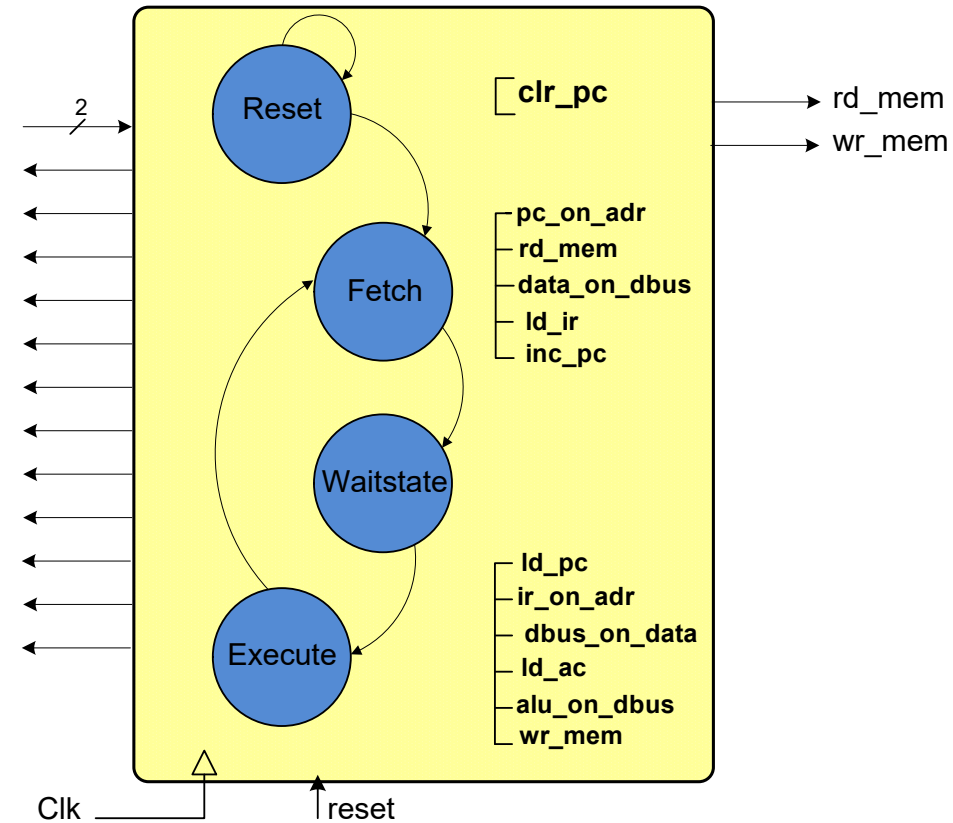
- What are buses?
- How are buses controlled?

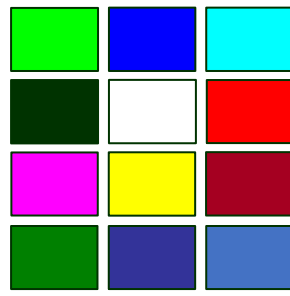
Intra – RTL Communication

- What are buses?
 - Simple wires
 - Shared wires
- How are buses controlled?



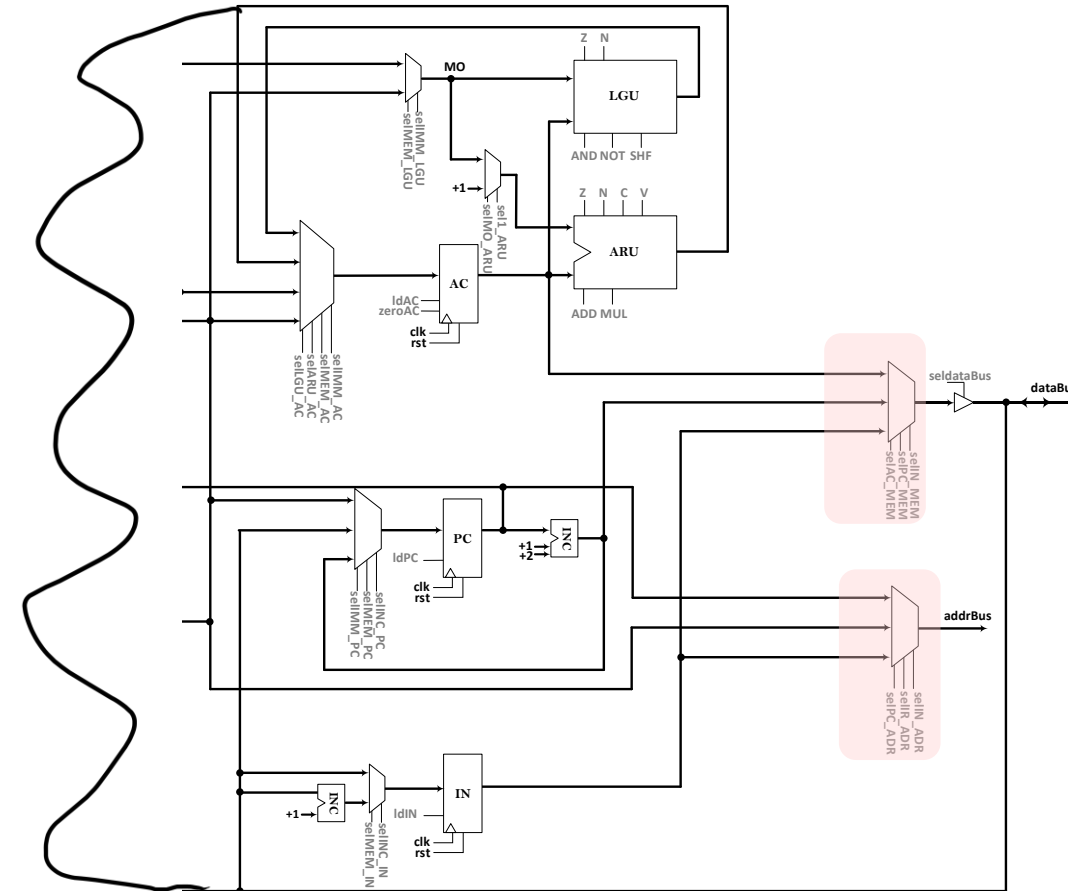
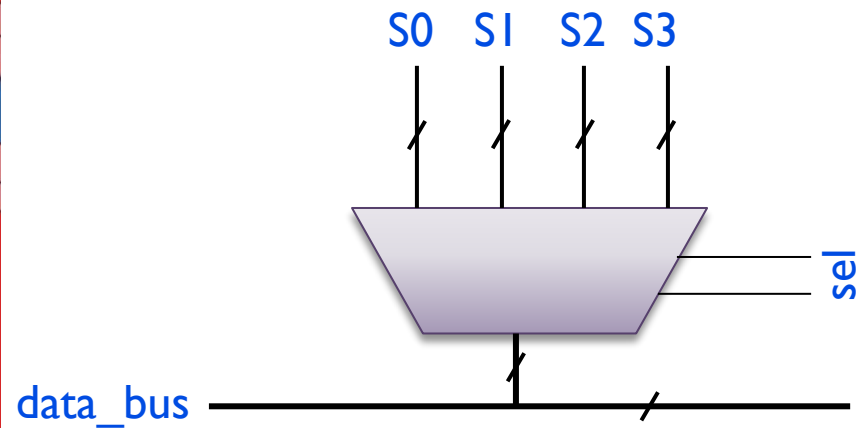
opcode
ir_on_adr
pc_on_adr
dbus_on_data
data_on_bus
ld_ir
ld_ac
ld_pc
inc_pc
clr_pc
pass
alu_on_dbus
add





Intra – RTL Communication

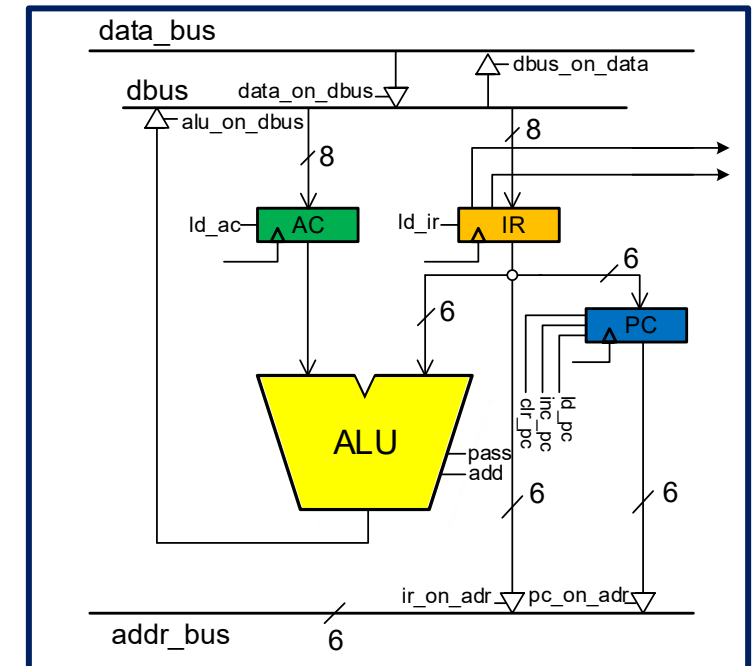
Multiplexed Bus



Intra – RTL Communication

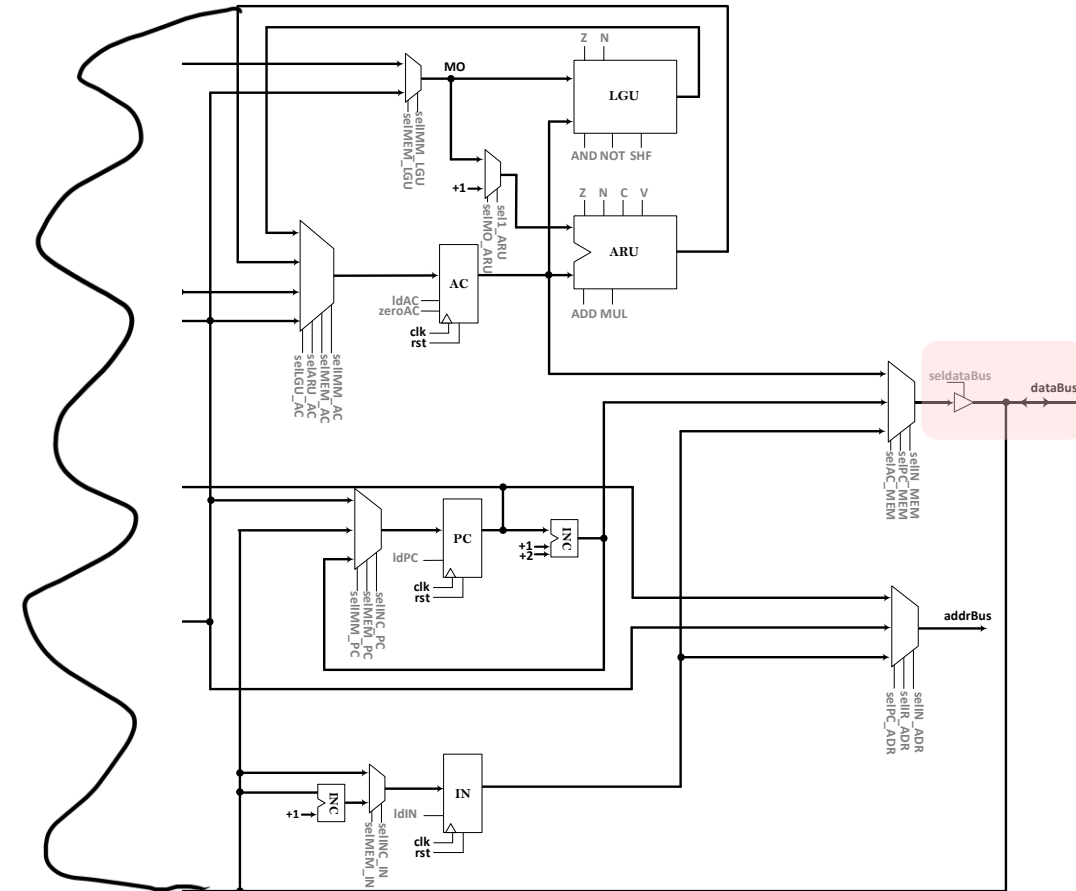
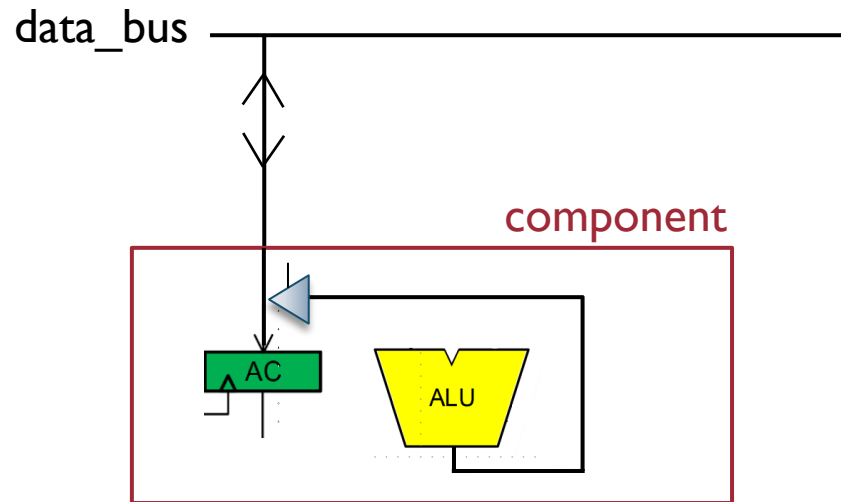
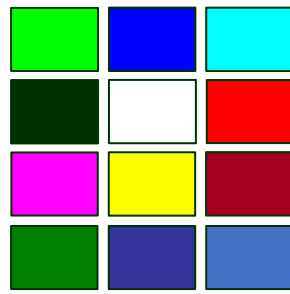
Three-State Bus

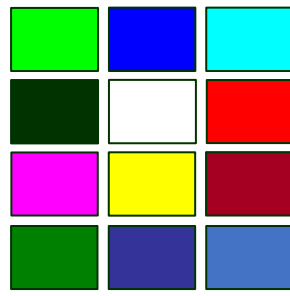
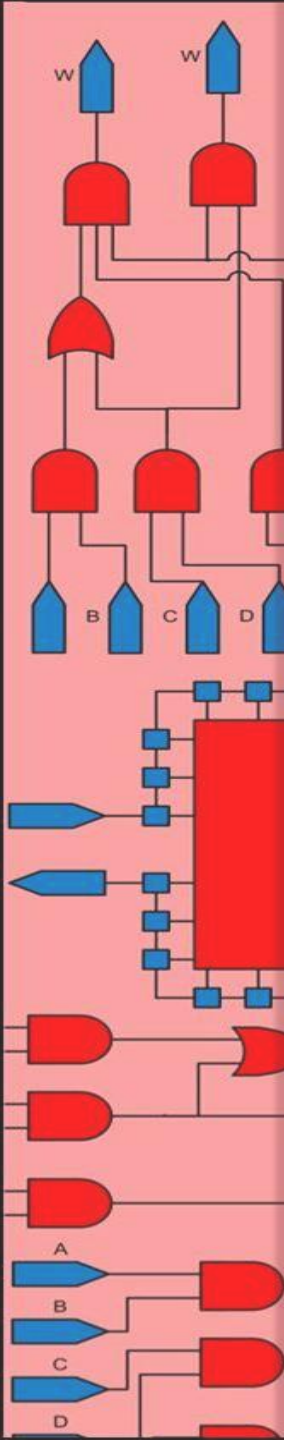
```
assign adr_bus = ir_on_adr ? ir_out[5:0] : 6'bzz_zzzz;  
assign adr_bus = pc_on_adr ? pc_out : 6'bzz_zzzz;  
assign dbus = alu_on_dbus ? alu_out : 8'bzzzz_zzzz;  
assign dbus = data_on_dbus ? data_bus : 8'bzzzz_zzzz;  
assign data_bus = dbus_on_data ? dbus : 8'bzzzz_zzzz;
```



Intra – RTL Communication

Bidirectional Bus





Outline

Intra – RTL Communication

Multiplexed Bus

Three State Bus

Bidirectional Bus

Inter – RTL Communication

What is handshaking?

Bus arbitration

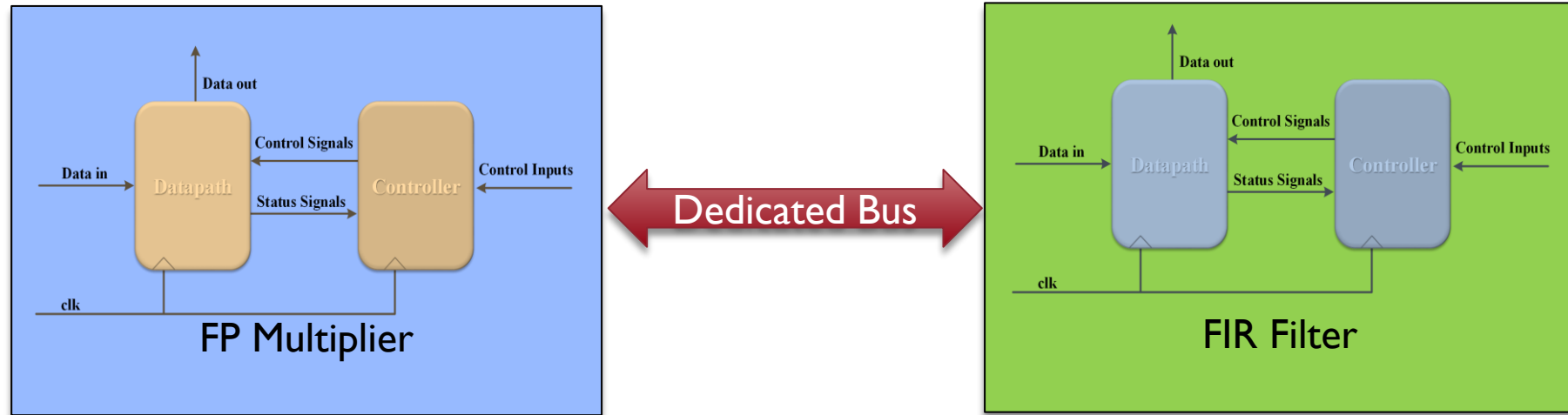
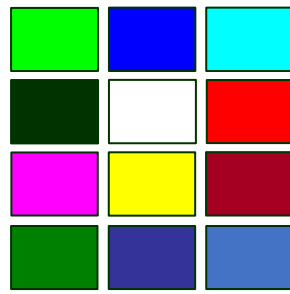
Device-to-device handshaking

Memory handshaking

IO wrappers

Inter – RTL Communication

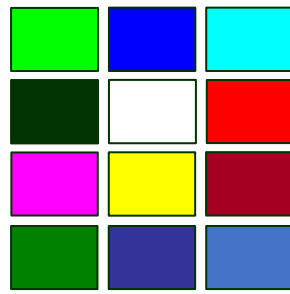
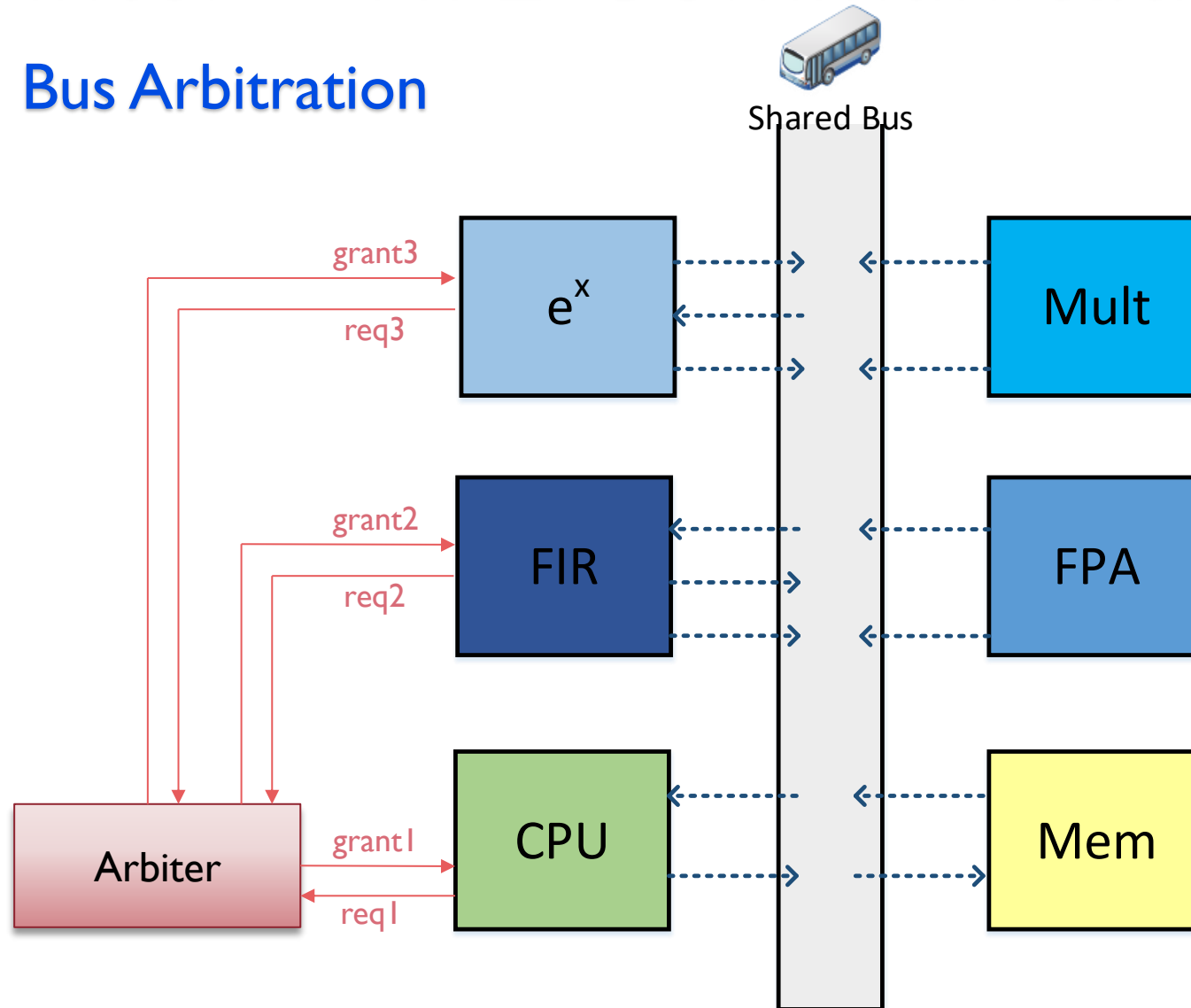
What is handshaking?



Dedicated
BUS

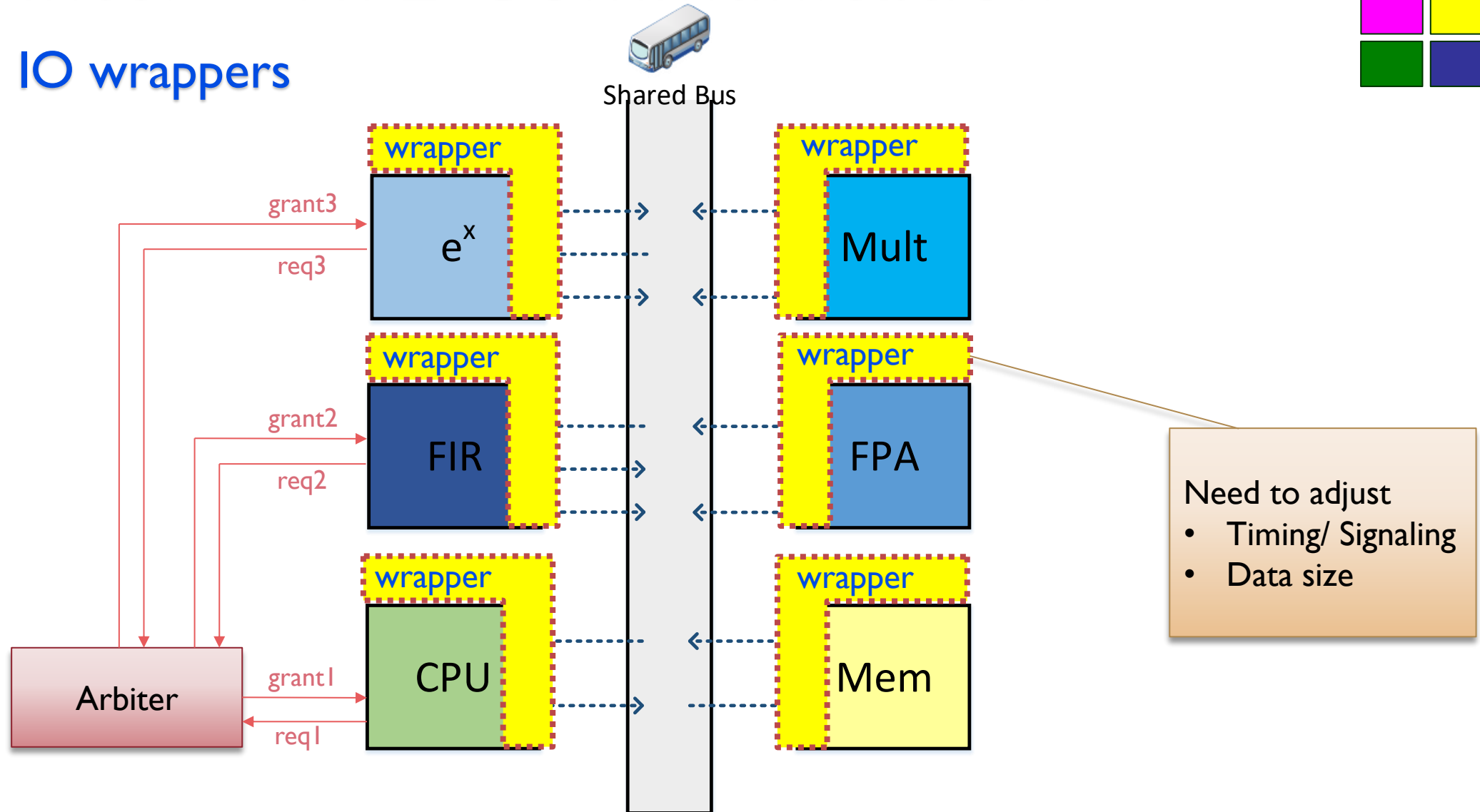
Inter – RTL Communication

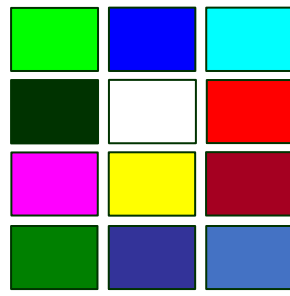
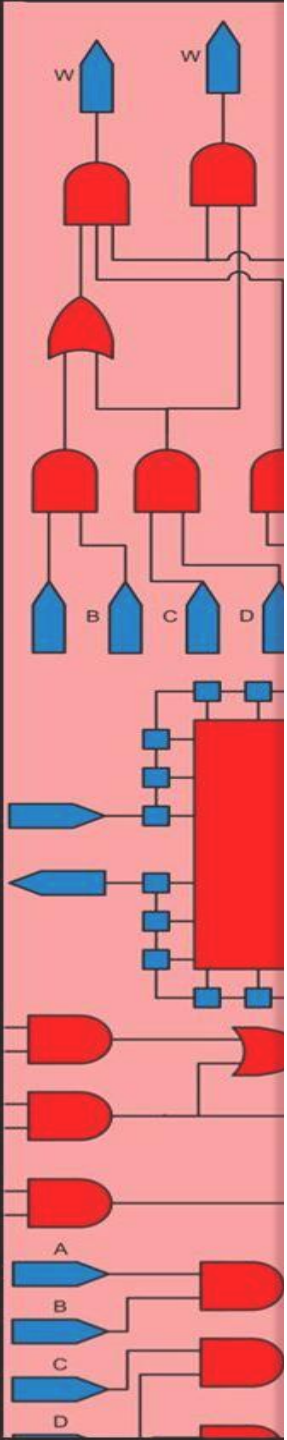
Bus Arbitration



Inter – RTL Communication

IO wrappers





Outline

Intra – RTL Communication

Multiplexed Bus

Three State Bus

Bidirectional Bus

Inter – RTL Communication

What is handshaking?

Bus arbitration

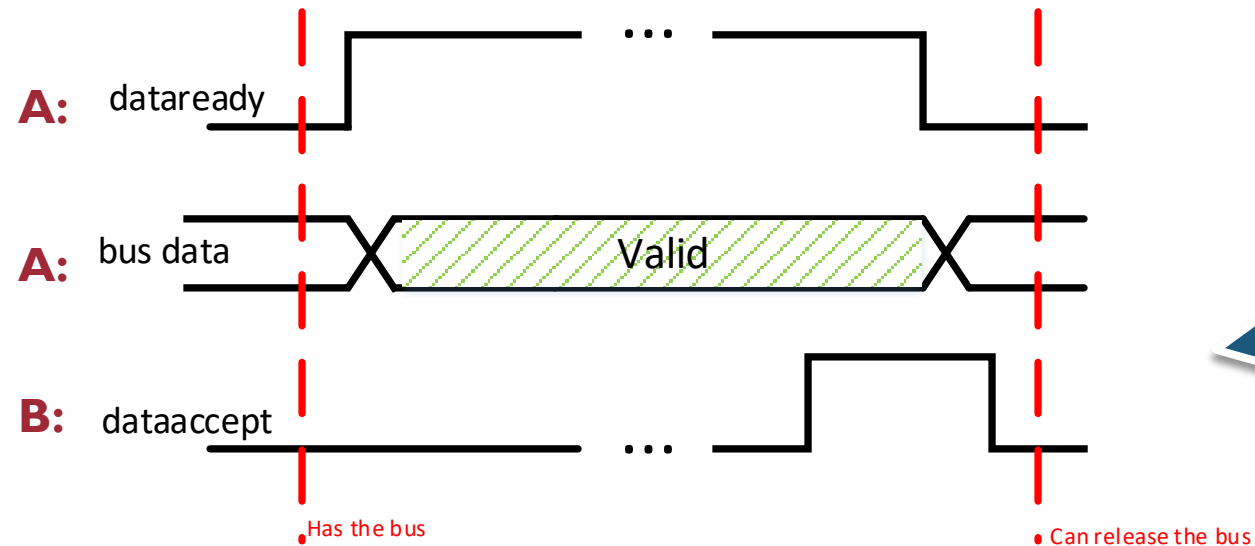
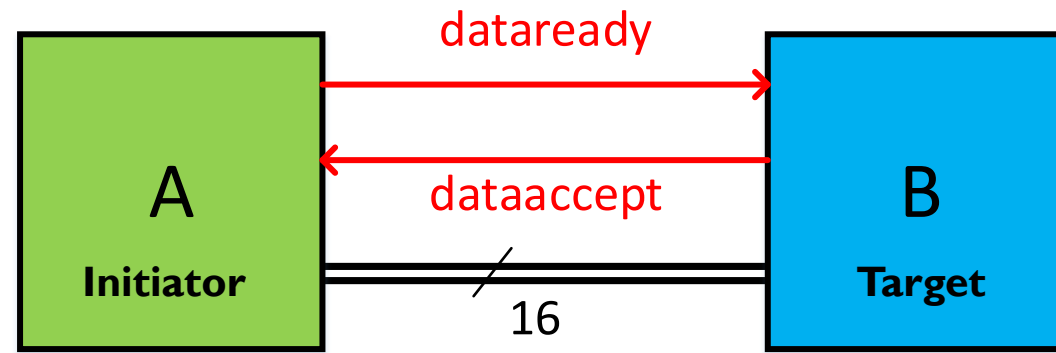
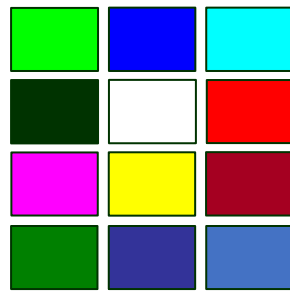
Device-to-device handshaking

Memory handshaking

IO wrappers

Handshaking

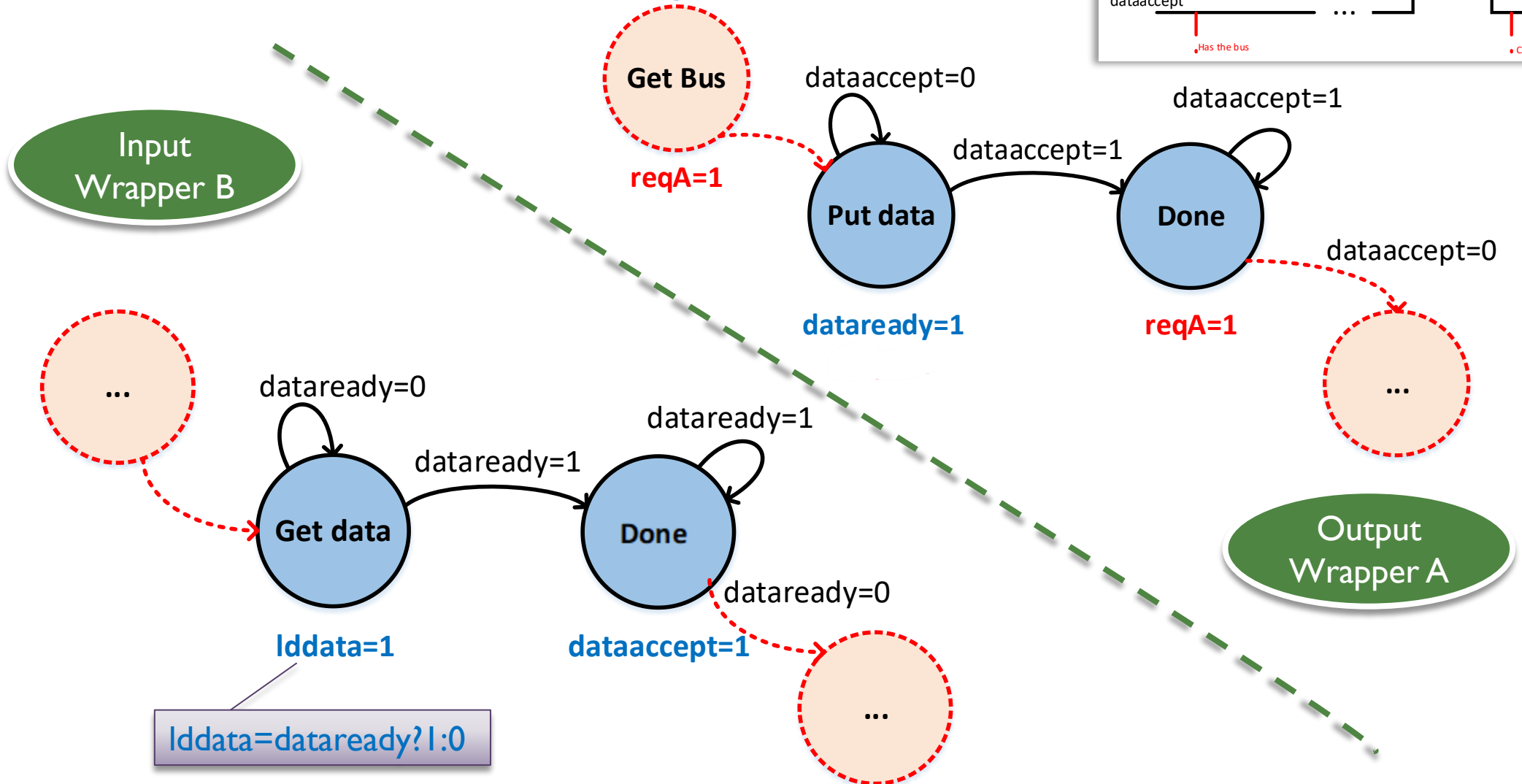
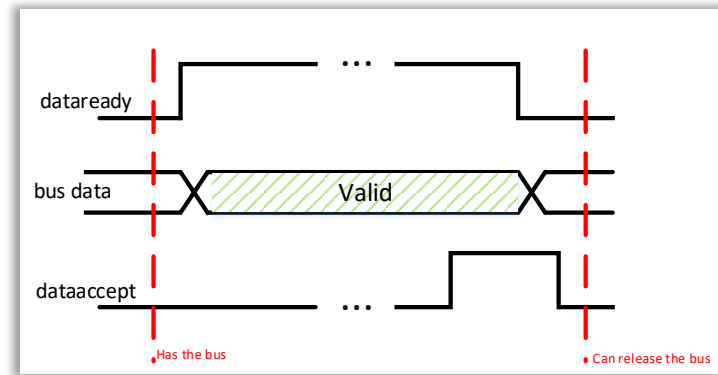
Device to Device Handshaking

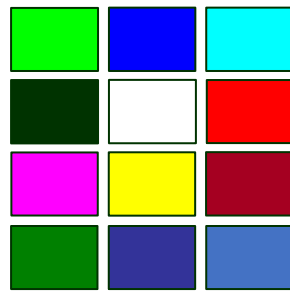


Fully Responsive
Two-line
Handshaking

Handshaking

Device-to-Device Handshaking





Outline

Intra – RTL Communication

Multiplexed Bus

Three State Bus

Bidirectional Bus

Inter – RTL Communication

What is handshaking?

Bus arbitration

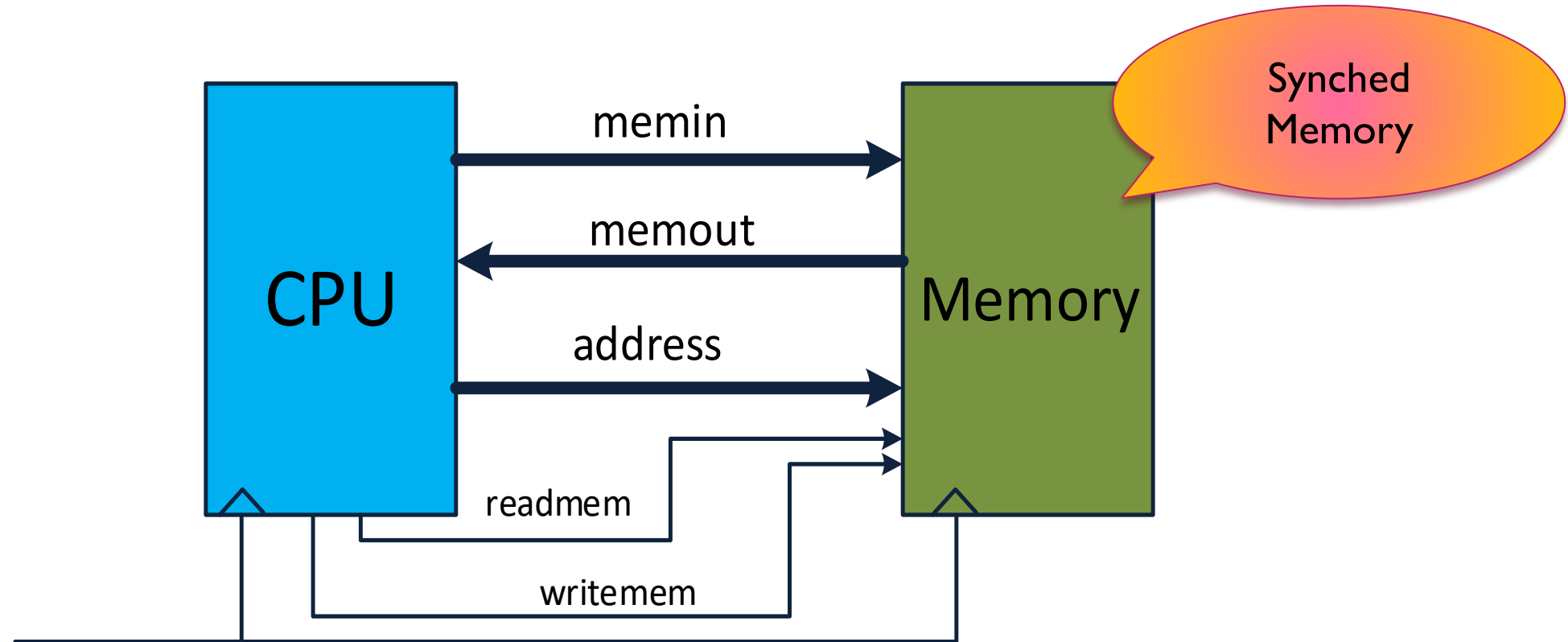
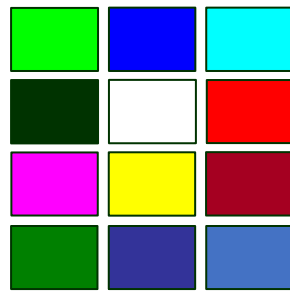
Device-to-device handshaking

Memory handshaking

IO wrappers

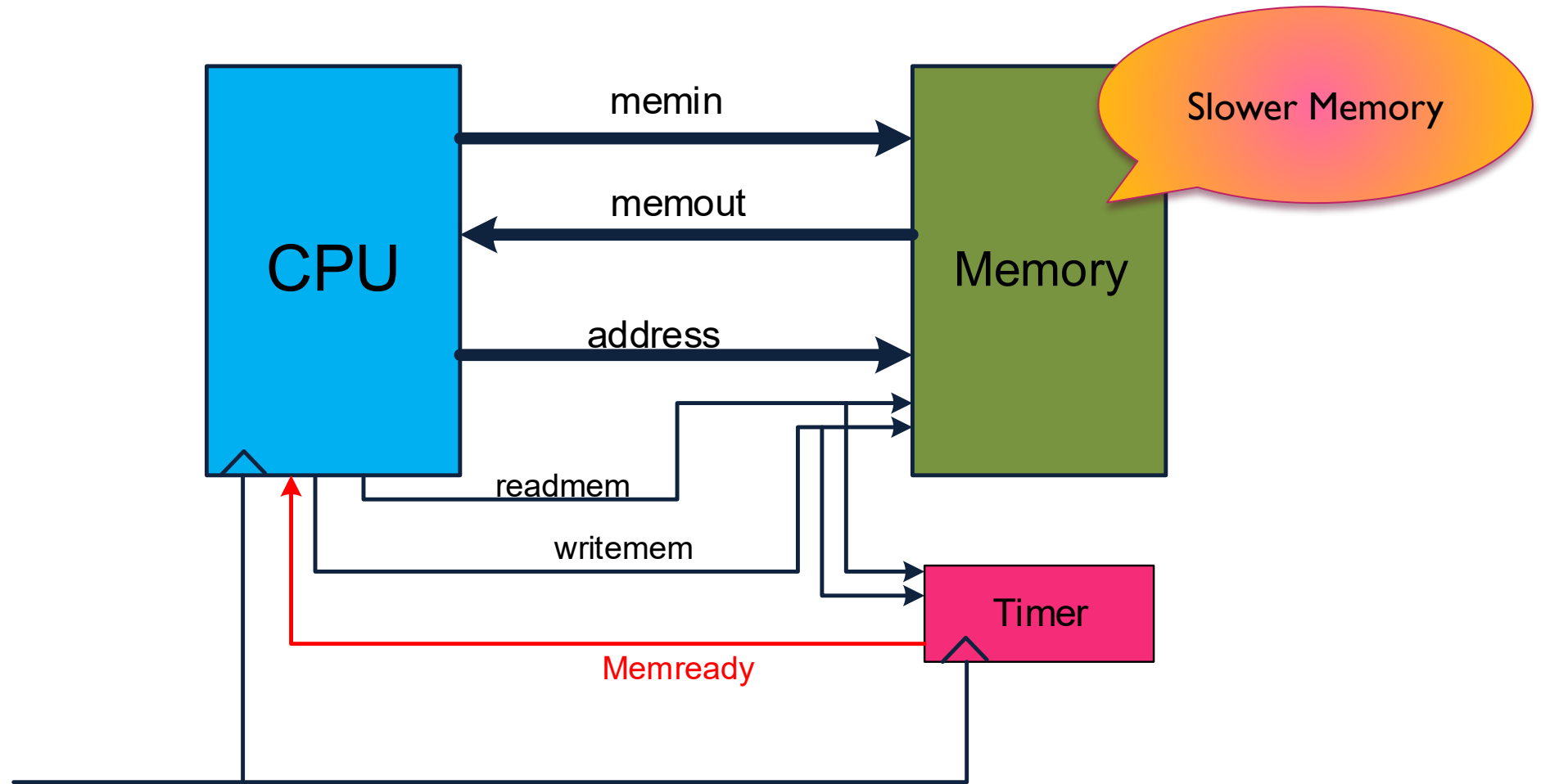
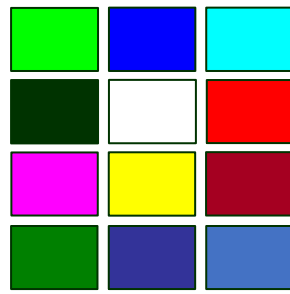
Handshaking

Memory Handshaking



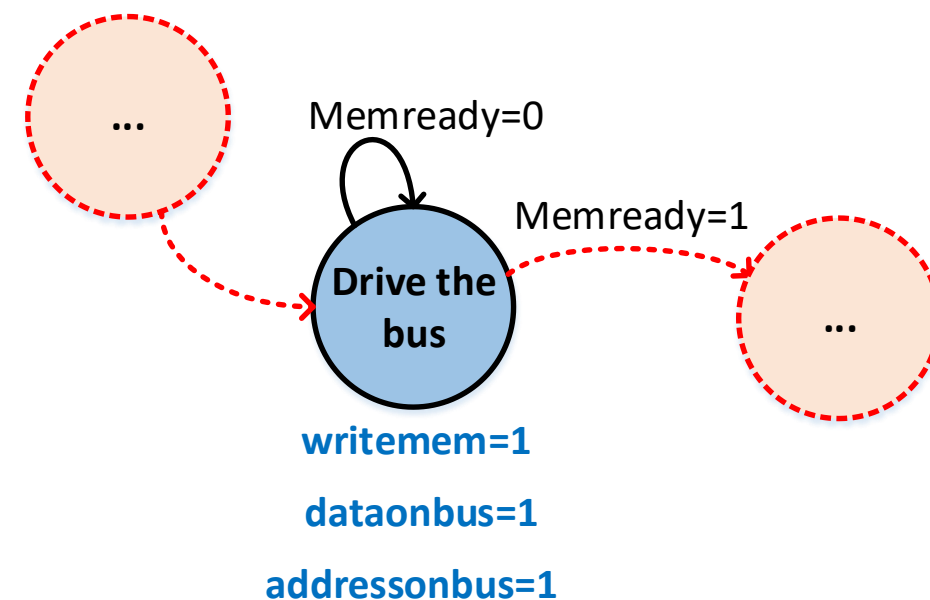
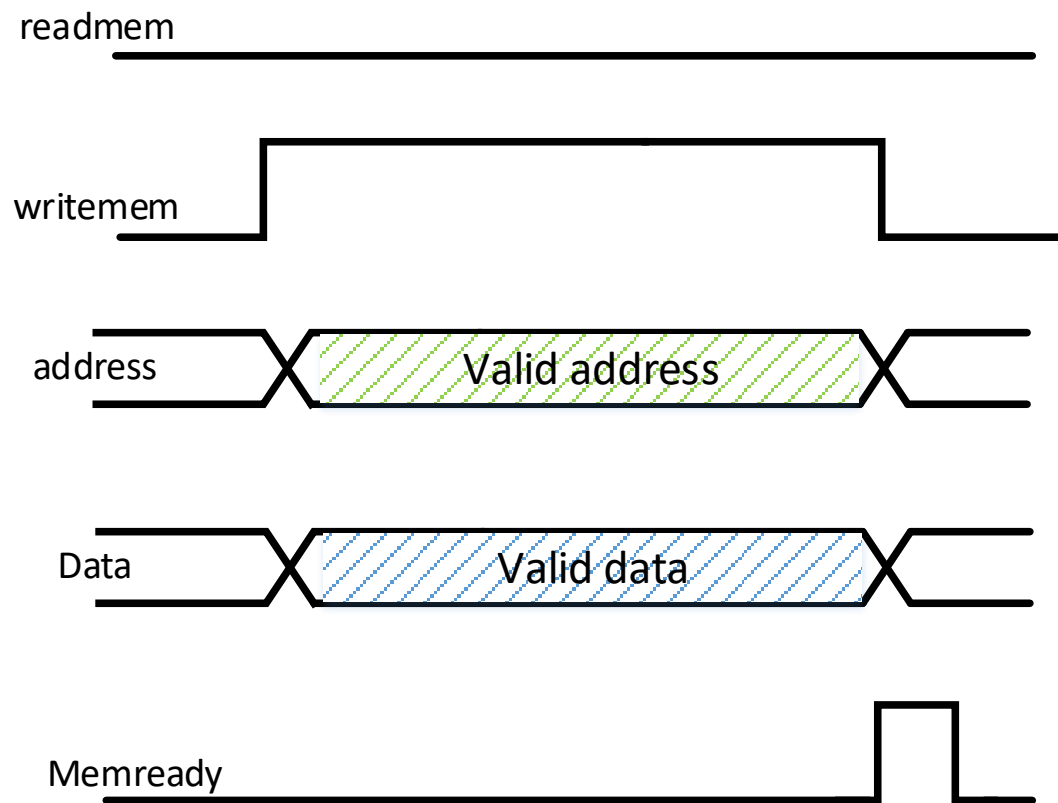
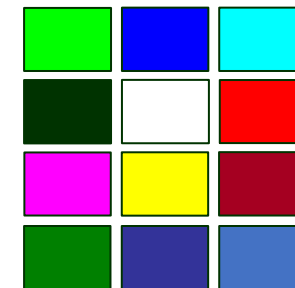
Handshaking

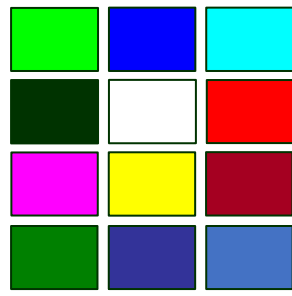
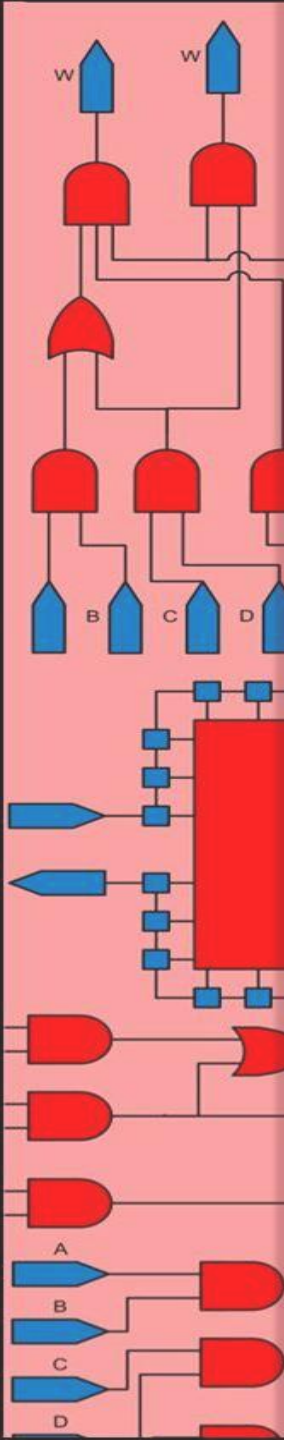
Memory Handshaking



Handshaking

Memory Handshaking





Outline

Intra – RTL Communication

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Three State Bus

Bidirectional Bus

Inter – RTL Communication

What is handshaking?

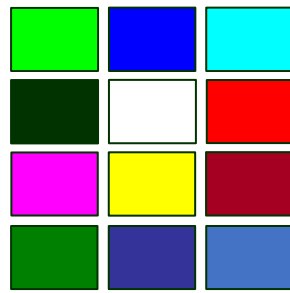
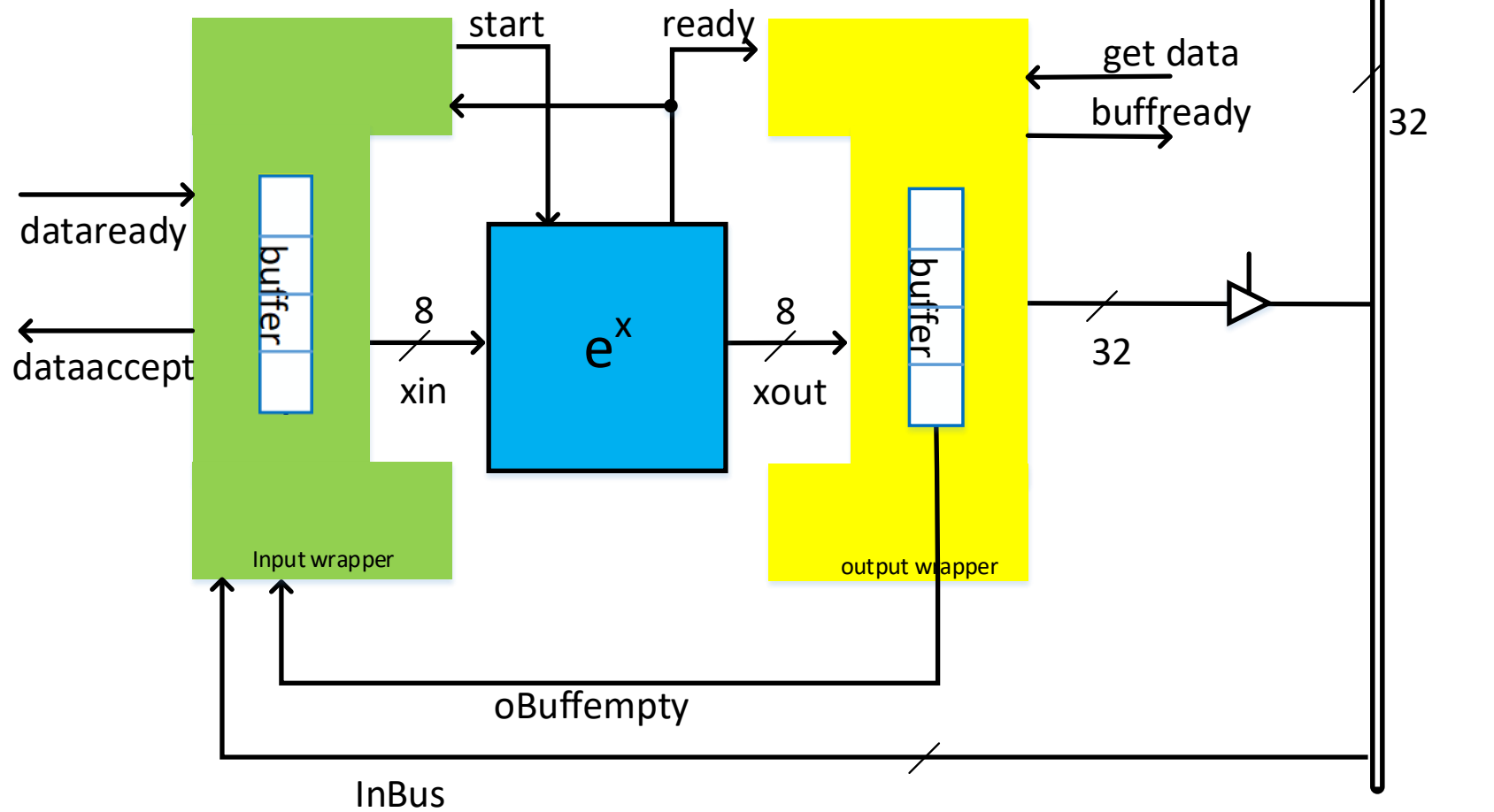
Bus arbitration

Device-to-device handshaking

Memory handshaking

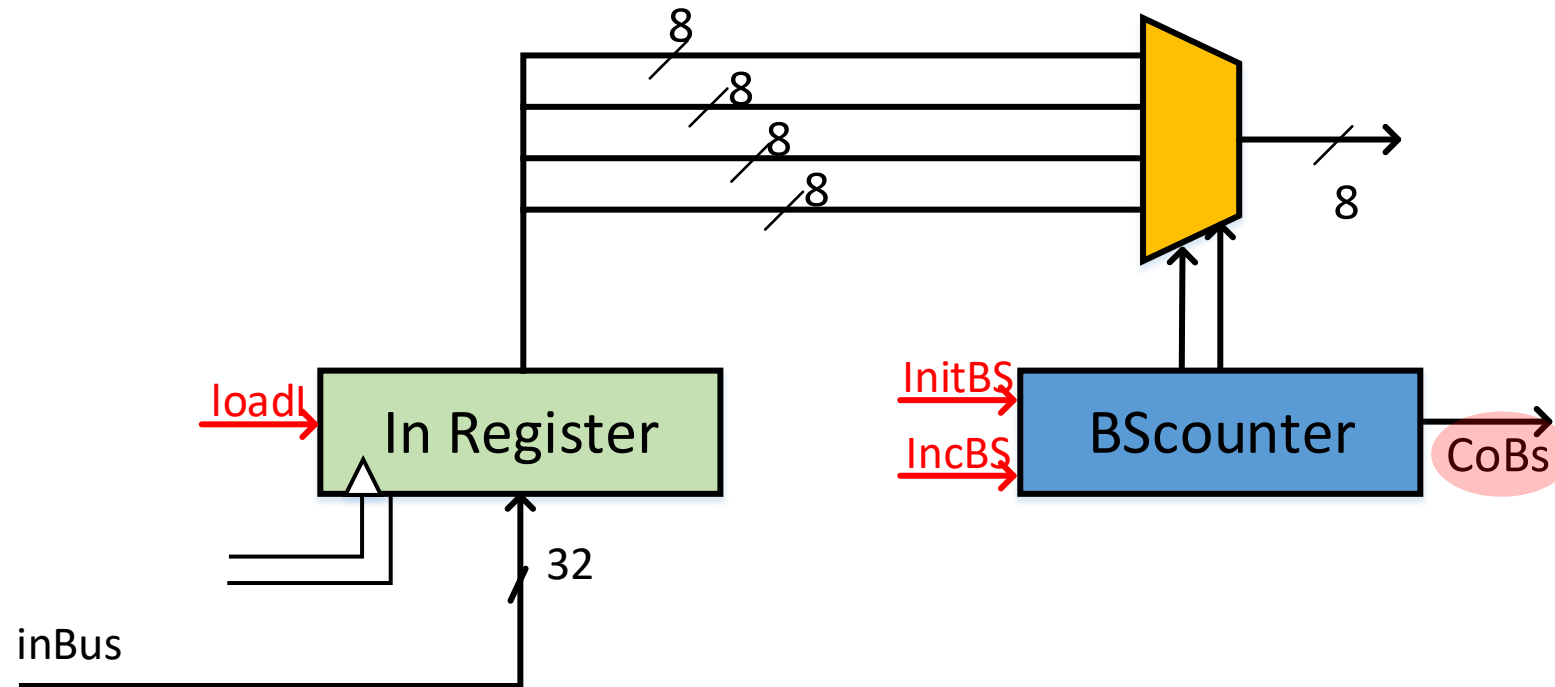
IO wrappers

IO wrappers



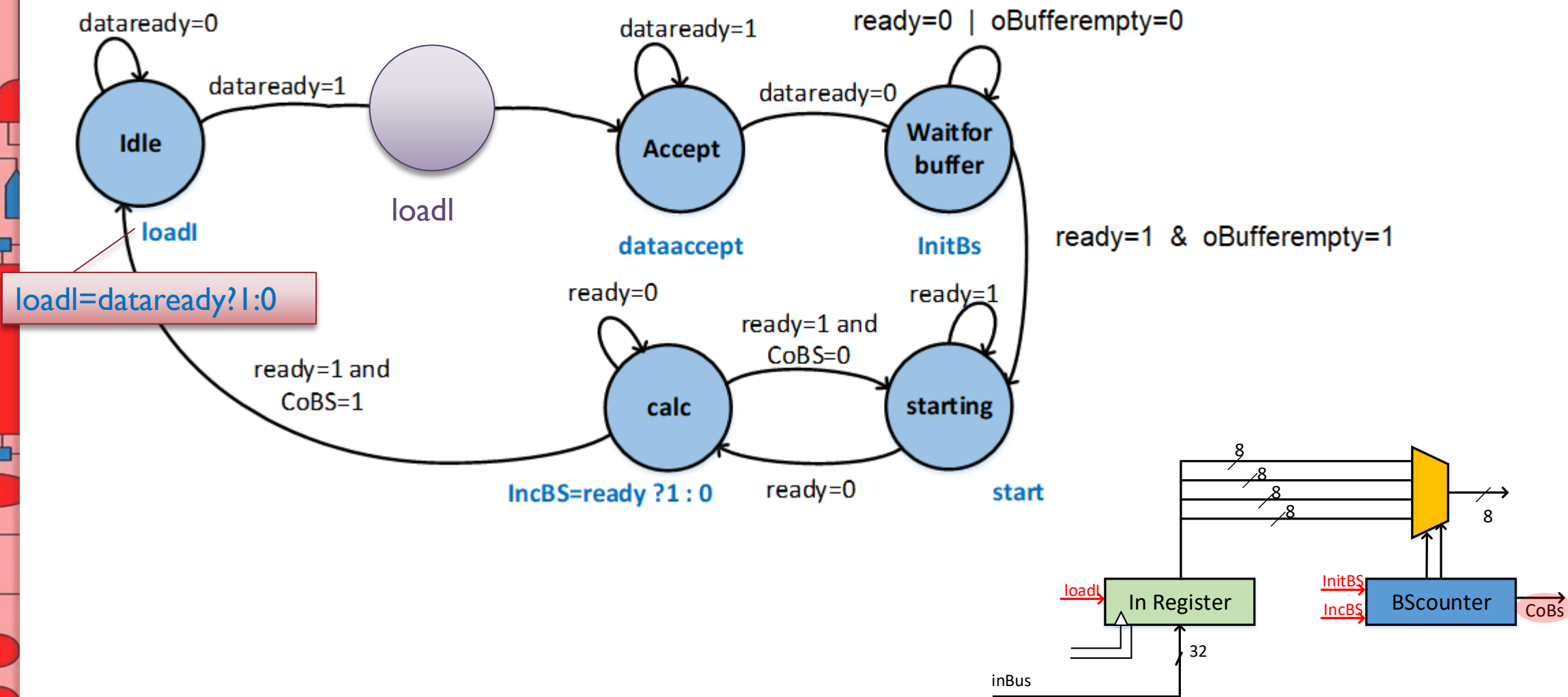
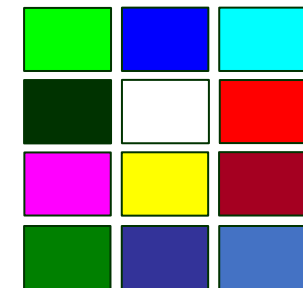
IO wrappers

Input wrapper - Datapath



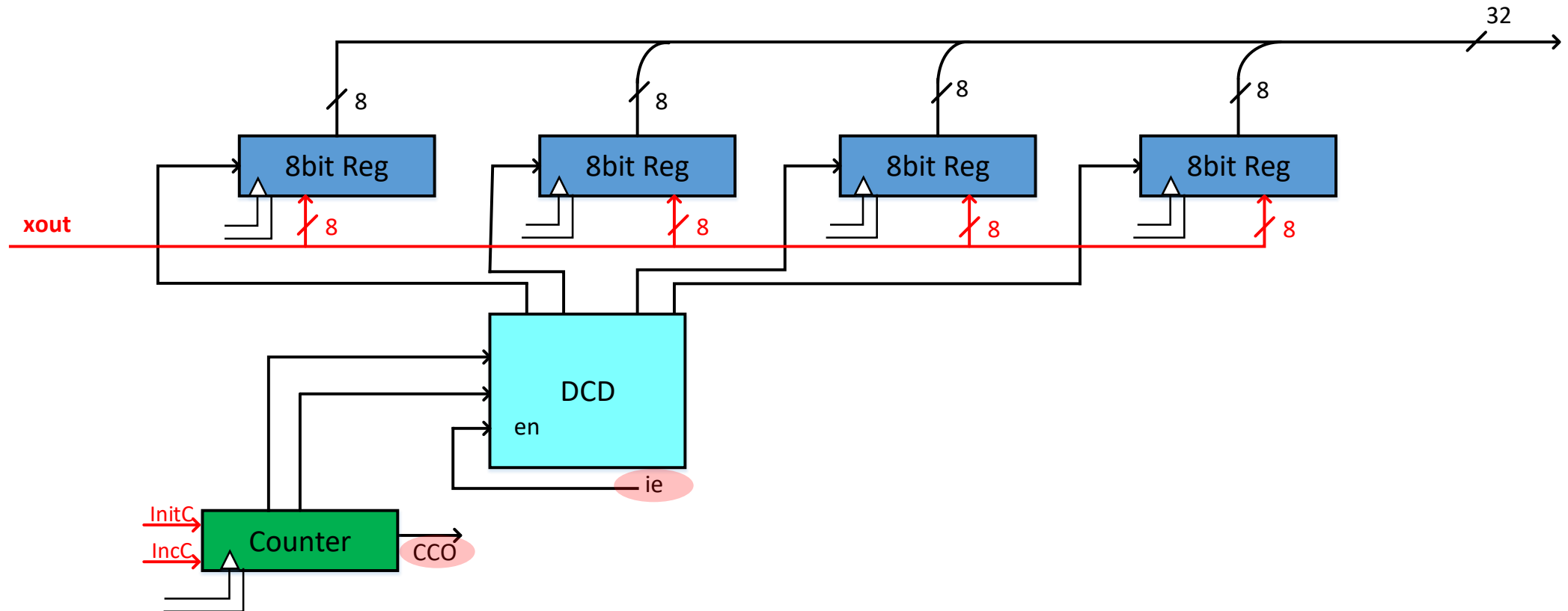
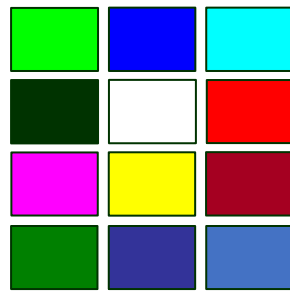
IO wrappers

Input wrapper - Controller



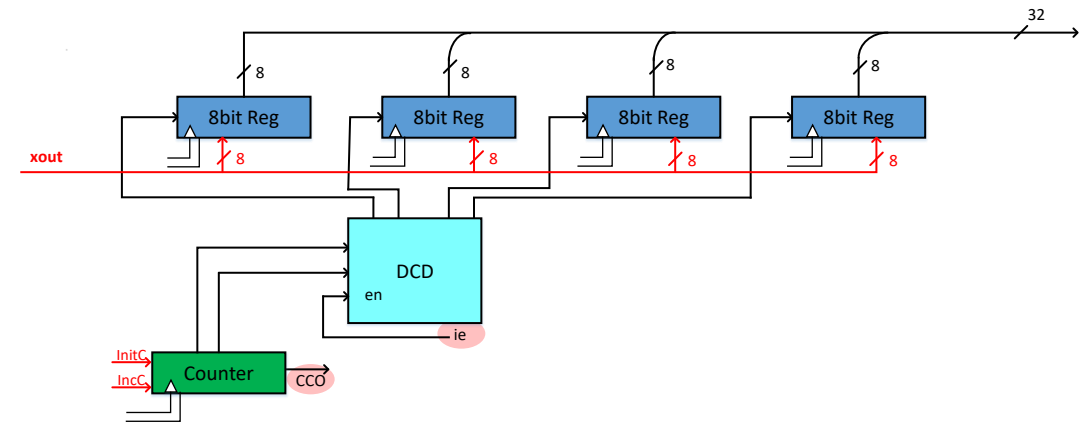
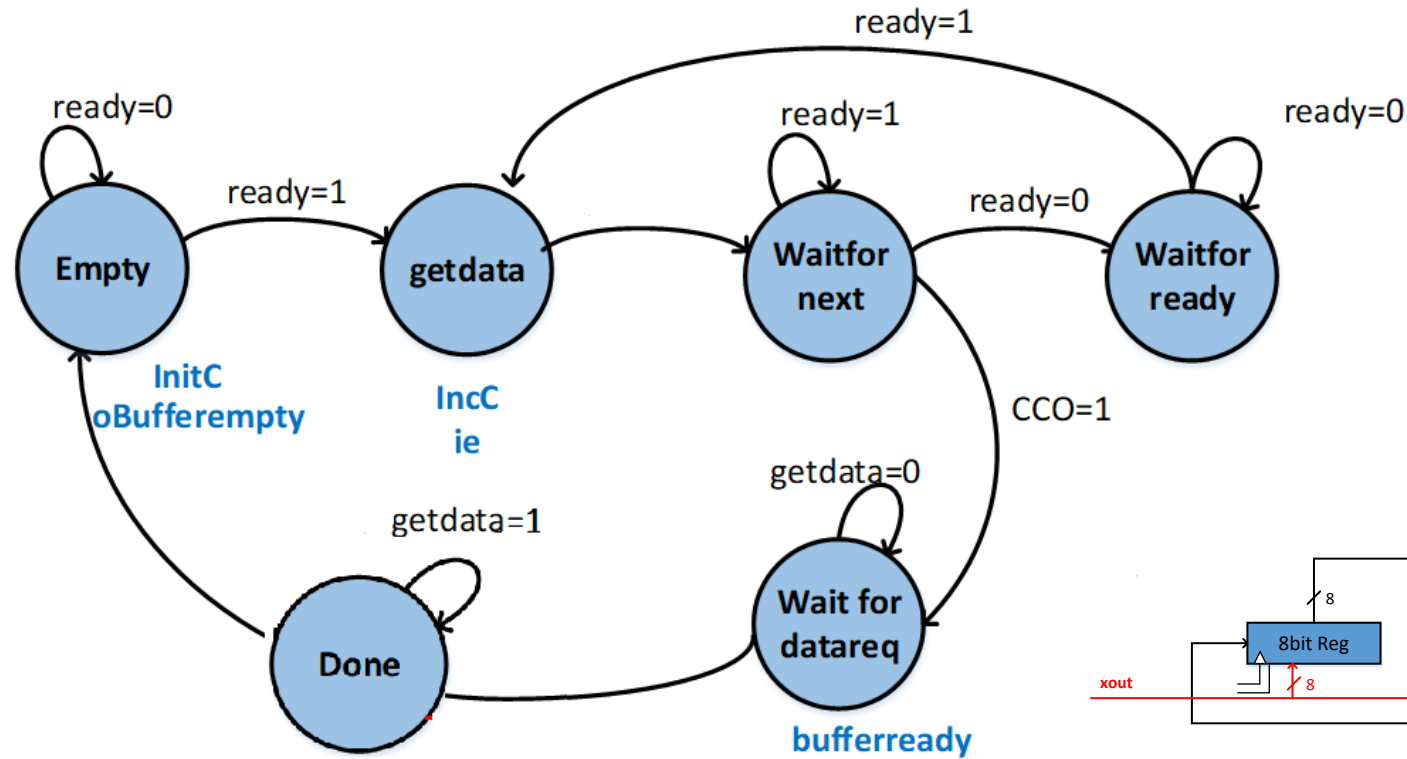
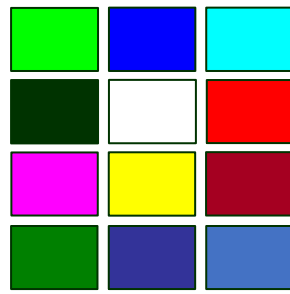
IO wrappers

Output wrapper - Datapath



IO wrappers

Output wrapper - Controller



Conclusion

We have covered these topics:

- Bussing
- Handshaking
- Bus arbitration
- Device to device handshaking
- Memory handshaking
- IO wrappers

